Customer Churn Prediction and Strategic Retention Engine

ConnectTel Communications Ltd.

Project Focus: End-to-End Machine Learning Deployment & API Development

Core Skills: Python | XGBoost | FastAPI | Feature Engineering | Imbalanced ML | Telecom Analytics

Outcome: Proactive churn detection engine with 76% recall and automated REST API.

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Executive Summary

ConnectTel Communications Ltd. is facing a critical business challenge: a steady rise in monthly customer churn, particularly among high-value postpaid users. Historically, the company's retention efforts were reactive—intervening only after a customer initiated a port-out request. This "late-stage" approach resulted in significant revenue loss and high customer acquisition costs.

This project transitions ConnectTel to a proactive retention model using a supervised machine learning engine. Using a dataset of 25,000 customers, we developed an optimized XGBoost Classifier. Recognizing that missing a churner is far more expensive than a false alarm, the model was heavily optimized for Recall. By engineering custom behavioral features—such as Technical Friction and Bill Shock—and adjusting the decision threshold to 0.42, the final model successfully captures 76% of all actual churners. The solution is finalized with a secured FastAPI prediction service, enabling real-time integration into ConnectTel's CRM.

1. Introduction & Project Motivation

1.1 The Telecom Landscape

The Indian telecommunications sector is one of the most competitive in the world. With the advent of 5G and fiber-to-the-home (FTTH), customer loyalty is no longer guaranteed by infrastructure alone. Customers have low switching costs and are frequently targeted by competitors.

1.2 The Problem of "Silent Churn"

ConnectTel identified that a large portion of churn was "silent." High-value customers were migrating to competitors without raising a single complaint, making it impossible for traditional customer service teams to intervene.

1.3 Project Motivation

The objective is to provide the Chief Commercial Officer (CCO) with a "radar system" to detect these silent churners through behavioral patterns hidden in billing, usage, and technical performance data.

2. Business Problem & Objectives

2.1 The Challenge

- Revenue Leakage: High-value customers migrating to competitors.
- Inefficient Marketing: Retention offers being sent to customers who were already loyal (waste of budget).
- Reactive Stance: Intervening too late in the customer lifecycle.

2.2 Project Objectives

- Predictive Modeling: Build an ML model to predict churn probability for individual subscribers.
- Driver Analysis: Understand the "Why" behind churn (e.g., technical friction vs. price dissatisfaction).
- Production Deployment: Create a real-time API (FastAPI) for CRM integration.
- Strategic Action: Design data-driven, tiered retention interventions.

3. Dataset Overview & Data Dictionary

The project utilized a rich dataset of 25,000 subscriber records with 36 attributes.

3.1 Data Categorization

- Demographics: Gender, Age, Region (Metro/North/South/West).
- Financial Metrics: Monthly Charges, Total Charges, ARPU (Average Revenue Per User).
- Usage Patterns: Avg Data GB, Voice Mins, SMS Count.
- Service Quality: Dropped Call Rate, Network Issues, NPS Score.
- Target: is_churn (1 = Churned, 0 = Retained).

3.2 Data Cleaning

- Missing Values: Coerced total_charges to numeric; filled missing values with the median.
- Identifier Removal: Dropped customer_id to prevent model overfitting.

4. System Architecture Pipeline

The project follows a modular production-grade pipeline designed for scalability:

[PLACEHOLDER: Insert Architecture Flowchart Diagram Here]

- Data Ingestion: Automated loading and validation of raw data.
- Preprocessing: Numerical scaling (StandardScaler) and categorical encoding (OneHotEncoder).
- Feature Engineering: Creating domain-specific interaction terms (Friction indices).
- Modeling: XGBoost training with class-weighting (scale_pos_weight).
- Deployment: Model serialization and API serving via FastAPI.

5. Exploratory Data Analysis (EDA) - Univariate

5.1 Target Variable Analysis

The dataset exhibits a natural class imbalance:

- Retained (0): ~70%
- Churners (1): ~30%

[PLACEHOLDER: Insert Plot 1 - Target Variable Distribution (Pie or Bar Chart)]

This imbalance is a critical finding, as it dictated the use of Precision-Recall metrics over standard Accuracy throughout the evaluation phase.

6. Exploratory Data Analysis (EDA) - Bivariate

6.1 Contractual Stickiness

"Month-to-Month" customers churn at 4x the rate of 2-year contract holders. This suggests that long-term contracts are the strongest anchor for retention.

[PLACEHOLDER: Insert Plot 2 - Churn Rate by Contract Type]

6.2 Service Quality Impact

There is a direct correlation between dropped_call_rate and churn. Customers with poor network experiences are significantly more likely to port out.

[PLACEHOLDER: Insert Plot 3 - Churn vs. Dropped Call Rate Boxplot]

7. Advanced Feature Engineering (The Friction Index)

Standard features were transformed into "Friction Indicators" to better capture the human reasons for leaving:

7.1 Technical Friction Index

- Formula: dropped_call_rate * (num_complaints_3m + 1)
- Business Logic: A dropped call is annoying, but a dropped call combined with unresolved complaints is a major churn trigger.

7.2 Bill Shock Signal

- Logic: A binary flag for users whose monthly_charges are in the top 25% of the distribution.
- Business Logic: Sudden price hikes or overage charges drive price-sensitive users away.

8. Advanced Feature Engineering (Value & Lifecycle)

8.1 Value Ratio

- Formula: arpu / (monthly_charges + 1)
- Business Logic: Measures perceived value; customers paying high fees but showing low usage feel they are overpaying.

8.2 Tenure Lifecycle Bucketing

- Categories: 0-1yr, 1-2yr, 2-4yr, 4-6yr, 6yr+.
- Business Logic: Allows the model to learn that a brand-new customer churns for different reasons than a long-term veteran.

9. Preprocessing & Feature Selection

9.1 Correlation Filtering

We performed a correlation analysis against the target variable to reduce noise. Only features with a correlation threshold > 0.05 were retained.

[PLACEHOLDER: Insert Plot 4 - Feature Correlation Heatmap]

9.2 The Pipeline

A ColumnTransformer was built to ensure consistent scaling and encoding across training and live API inference.

10. Theoretical Framework: XGBoost

10.1 Why XGBoost?

XGBoost (eXtreme Gradient Boosting) was selected as the core engine because:

- Non-linearity: It handles complex, non-linear relationships expertly.
- Regularization: Built-in L1/L2 regularization prevents overfitting.
- Imbalance Handling: It allows for custom weights on the minority class.

11. Training & Hyperparameters

11.1 Class Imbalance Strategy

We used a scale_pos_weight of 1.4137. This forces the algorithm to penalize errors on churners more heavily than errors on non-churners.

11.2 Optimal Parameters

- n_estimators: 100
- learning_rate: 0.05 (Ensures slow, robust learning)
- max_depth: 4 (Prevents the model from memorizing noise)
- subsample: 0.7

12. Evaluation & Threshold Optimization

12.1 The Decision Policy

In telecom retention, a False Negative (missing a churner) is much more expensive than a False Positive (wrongly flagging a loyal customer).

12.2 Threshold Shifting

The decision threshold was shifted to 0.42. This aggressively maximizes Recall, ensuring ConnectTel's "radar" catches as many churners as possible.

13. Final Performance Metrics

Evaluated on a 20% holdout test set (5,000 records), the optimized model achieved:

- Final AUC-ROC Score: 0.6564
- Recall (Class 1): 76% (Successfully identifies 3 out of 4 departing customers).
- Precision (Class 1): 0.49
- F1-Score: 0.60

[PLACEHOLDER: Insert Plot 5 - ROC Curve]

[PLACEHOLDER: Insert Plot 6 - Confusion Matrix]

14. Feature Importance Analysis

The model identified the following top drivers for churn at ConnectTel:

1. Contract Type (Strongest predictor)

2. Technical Friction (Network Quality)

3. Tenure Bucket (Customer Lifecycle)

4. Monthly Charges (Pricing)

[PLACEHOLDER: Insert Plot 7 - XGBoost Feature Importance Bar Chart]

15. Production Deployment Architecture

The model was serialized and deployed as a RESTful API using FastAPI.

[PLACEHOLDER: Insert REST API Architecture Diagram]

15.1 API Capabilities

- Endpoint /predict: Real-time scoring for individual customer agents.
- Endpoint /predict_batch: Processes bulk CSV uploads for monthly marketing campaigns.
- Security: Authenticated via X-API-Key to protect customer privacy.

16. Business Strategy & Retention Matrix

Based on the 76% Recall rate, we recommend a Tiered Retention Strategy:

1. HIGH RISK + HIGH VALUE: Proactive Call / Personalized High-Value Offer (Cost: High)

2. HIGH RISK + MEDIUM VALUE: Automated App Notification / Bonus Data Pack (Cost: Low)

3. LOW RISK + ANY VALUE: No Intervention / Standard Marketing (Cost: Zero)

17. Repository Structure & Deployment

17.1 Folder Structure

- /app: FastAPI logic and endpoints.
- /models: Optimized .pkl model artifact.
- /notebooks: Detailed EDA and Training logs.
- /data: Raw and processed dataset versions.

17.2 Installation

```
git clone https://github.com/abhishekankushe/Capstone-Project.git

pip install -r requirements.txt

uvicorn app.main:app --reload
```

18. Conclusion & Future Scope

18.1 Conclusion

This project provides ConnectTel with a 76% sensitive "radar" for customer churn. By moving from reactive to proactive analytics, the company can now protect its revenue base more efficiently.

18.2 Future Roadmap

- SHAP Explainability: Adding "Reason Codes" to every prediction.
- Dockerization: Deploying the API as a container for cloud scale.
- A/B Testing: Measuring the conversion rate of the new retention offers.